FELIX SPRINGER

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Germany ◇ Augsburg

EDUCATION

Universität Augsburg October 2023 - August 2024

Bachelor of Science: Mathematics

I just did this for fun and did not pursue a degree

Leibniz Universität Hannover October 2016 - April 2021

Bachelor of Science: Physics

Major subject: Physics grade point average: 2.4
Minor subject: Computer Science

Thesis: "Storage Register Design for an Ion Trap Quantum Processor"

Viktoria-Luise-Gymnasium Hameln July 2008 - June 2016

General qualification for university entrance

Advanced courses: Physics, Mathematics, Geography grade point average: 2.2

WORK EXPERIENCE

Natuvion GmbH September 2024 - today

Software Developer

- Development of a domain-specific language in Haskell, targeting Dhall, to create an abstraction over ERP system transformations
- Reverse-engineering the behavior of a mature ABAP codebase to formalize its semantics and improve correctness
- Spearheaded the adoption of the *deep-transformations* library by annotating every node of the AST using functors, allowing us to write folds and traversals over the structure of the AST

Possehl Analytics GmbH April 2021 - September 2023

Software Developer

- DevOps using Nix, Docker, git, AWS, Hetzner
- Backend-Programming in Haskell: servant, aeson, stm, mtl, megaparsec, wai, exceptions, ...
- Database rollout and maintenance: PostgreSQL, MongoDB
- Design and implementation of a customer-independent master data API

LUH: Institut für Quantenoptik February 2020 - March 2021

Technical Supervisor

- Operation, installation and maintenance of media technologies
- Recording and editing videos of experiments for Physics lectures
- Event management

LUH: Institut für Botanik March 2019 - June 2020

System Administrator

- Administration of the network and local servers
- Maintaining and providing personal computers
- Automating and monitoring data backup

LUH: Institut für Angewandte Mathematik October 2018 - February 2019

Tutor for the lecture "Mathematics 1 for Life Science and Earth Sciences"

- Weekly tutoring a class according to the lecture
- Grading exercises and exams

PERSONAL PROJECTS

mensam (feature-complete but stale) github.com/jumper149/mensam

is a *desk booking web application* for coworking spaces and offices. The backend is built with *Haskell* and the frontend is written in *Elm*. Everything is tied together with *Nix*. You can try out mens.am.

homepage (actively used and maintained) github.com/jumper149/homepage

is an *HTTP server*, that focuses on configurability. I am using this *Haskell* project for my personal homepage. It includes a Blog and *Atom Feed*, that is generated from *AsciiDoc*. The effect system is based on *mtl*, *monad-control* and a composable transformer stack. It provides a *NixOS module* via a *flake*.

go (discontinued) github.com/jumper149/go

is a strategy board game and this implementation extends the ruleset by providing different boards to play on. This is a *fullstack* project with an *HTTP server* and a *frontend web application*, that is compiled with *GHCJS* and uses a *WebSocket* to communicate with the server. It's written in *Haskell*, making use of various *extensions* to the type systems. It builds with *Nix* and integrates well with *NixOS*.

blugon (still maintained) github.com/jumper149/blugon

is a simple and configurable Blue Light Filter for *X11*. It's written mostly in *Python* and the main focus is to follow **nix* standards.

TECHNICAL SKILLS

Languages	German, English
Programming languages	Haskell, Idris, Bash, Nix, Python, C, Agda, Wolfram Mathematica, Scheme
Software/Tools	GNU Coreutils, Linux, Git, Nix, Vim, SSH, tmux, LaTeX, AsciiDoc, SQL, HTML, CSS, POSIX, HTTP